

CURATED INTERMEDIATE WORKSHEET

INTEGERS: BODMAS SIMPLIFICATION (1D) & APPLIED WORD PROBLEMS (1E)

Key Operational Principles

BODMAS Rule for Order of Operations: Always simplify in the strict order: Brackets in hierarchy: Bar/Vinculum ($\overline{x-y}$), Round (), Curly { }, then Square []. **O**f (Multiplication). **D**ivision ($\div$) and **M**ultiplication ($\times$) from left to right. **A**ddition (+) and **S**ubtraction ($-$) from left to right.

Real-World Signed Translations: Upward motion, deposits, gains, heating, and East are positive (+). Downward motion, withdrawals, losses, cooling, and West are negative (-).

SECTION A: MULTIPLE CHOICE QUESTIONS (10 MARKS)

Q1. Simplify the expression: $18 - [14 - \{15 - (8 - \overline{7 - 3})\}]$.

- (a) 15
- (b) 7
- (c) 11
- (d) 19

Q2. If $x = -12 \div \{(-3) + 1\}$ and $y = (-15 \div -3) - (-2)$, what is the value of $x \times y$?

- (a) 42
- (b) -42
- (c) 24
- (d) -24

Q3. Simplify the expression: $(-50) \div \{15 - (-5) \times (-2)\} \times (-4)$.

- (a) 40
- (b) -40
- (c) 250
- (d) -250

Q4. A research submarine at a depth of -450 m dives further down at a rate of 15 meters per minute for 12 minutes. What is its new depth coordinate?

- (a) -630 m
- (b) -270 m
- (c) -465 m
- (d) -530 m

Q5. In an international math contest, $+4$ marks are awarded for a correct answer, and -3 marks are given for an incorrect answer. If Shreya scored -5 marks and got 7 correct answers, how many questions did she answer incorrectly?

- (a) 11

- (b) 8
- (c) 9
- (d) 12

Q6. A merchant earns a profit of Rs. 5 on each notebook sold and incurs a loss of Rs. 2 on each pen sold. In a given month, she makes an overall loss of Rs. 10. If she sold 16 notebooks, how many pens did she sell?

- (a) 45
- (b) 35
- (c) 40
- (d) 50

Q7. Evaluate: $-24 + [-18 \div \{12 \div (4 - \overline{2 - 4})\}]$.

- (a) -33
- (b) -15
- (c) -21
- (d) -27

Q8. The freezing point of a chemical solution is -18°C . When an additive is mixed in, the freezing point drops by 3°C every minute. What will the solution's freezing point be after 7 minutes?

- (a) -39°C
- (b) -3°C
- (c) -25°C
- (d) -45°C

HOTS Question 9

Define a binary operation $\otimes$ for any integers a and b as: $a \otimes b = (a \times b) - (a + b) \div (-1)$. If $x = -4$ and $y = 3$, evaluate $x \otimes y$.

- Q9.**
- (a) -13
 - (b) -11
 - (c) -15
 - (d) -7

HOTS Question 10

In the expression $B - \{C \times (D - \overline{E - F})\}$, let $B = -6$, $C = 2$, $D = -1$, and $F = 5$. If the entire expression evaluates to -12, find the value of the integer E .

- Q10.**
- (a) 1
 - (b) 2
 - (c) -1
 - (d) 0

SECTION B: SUBJECTIVE PROBLEMS (30 MARKS)

P1. Order of Operations: Simplify the following expression using BODMAS:

$$(-15) \times (-4) - [-25 \div \{10 - (3 - 4 - 6)\}]$$

P2. Nested Brackets and Vinculum: Simplify the following:

$$-36 + [-48 \div \{-12 + 2 \times (15 - \overline{12 - 5})\}]$$

P3. Absolute Value and Division: Evaluate the following expression step-by-step:

$$|-15 - (-3)| \times \{(-24) \div (-6)\} - [-40 \div \{13 + (-3) \times 3\}]$$

P4. Mountain Temperature Ascent: The temperature at the base of a high mountain is 15°C . As an explorer climbs, the temperature decreases by 2°C for every 300 meters of ascent. If the explorer climbs to an altitude of 2400 meters and then descends 600 meters to set up camp, what is the temperature at the campsite?

P5. Corporate Account Ledger: A small tech firm's bank balance was Rs. 12,500 at the start of the week. During the week:

- They deposited 3 client checks of Rs. 4,200 each.
- They withdrew Rs. 15,600 to pay operational salaries.
- A penalty fee of Rs. 150 was debited for maintaining a balance below a threshold mid-week.
- Finally, they returned a defective purchase, resulting in a credit refund of Rs. 1,850.

Determine the company's final bank balance. Show step-by-step signed integer additions and subtractions.

P6. Hovering Altitude and Underwater Coordinates: A search-and-rescue helicopter is hovering at an altitude of 1250 meters above sea level. Directly below it, a submarine is submerged at a depth of 450 meters below sea level.

- (a) Find the vertical distance between the helicopter and the submarine.
- (b) If the helicopter climbs an additional 150 meters and the submarine dives a further 120 meters, find the new vertical distance between them.

P7. Academic Competition Scoring: In a school math tournament, there are 40 questions. +5 marks are awarded for each correct answer, -2 marks are given for each incorrect answer, and -1 mark is deducted for each unattempted question. Ananya attempted 32 questions and left the rest unattempted. If she scored a net total of 110 marks, how many questions did she answer correctly and how many incorrectly?

P8. Commercial Microchips Portfolio: A wholesale dealer of electronic microchips earns a profit of Rs. 120 on each Grade-A chip sold, but incurs a loss of Rs. 45 on each Grade-B chip sold.

- (a) In a particular week, the dealer sold 450 Grade-A chips and some Grade-B chips. If his net profit for the week was Rs. 36,000, find the number of Grade-B chips sold.
- (b) If in the following week, the dealer wants to make a net profit of at least Rs. 45,000 and expects to sell 600 Grade-B chips, what is the minimum number of Grade-A chips he must sell to meet this goal?

HOTS Problem 9

Consider the custom function $f(x, y)$ defined for integers x and y as:

$$f(x, y) = [x \div \{y - (x + \overline{y - 2})\}]$$

Find all pairs of integers (x, y) such that $-3 \leq x \leq 3$, $1 \leq y \leq 4$, and the function $f(x, y)$ is defined and evaluates to an integer. Simplify the expression inside the curly brackets first to find a general algebraic shortcut.

P9.**HOTS Problem 10**

A municipal water reservoir has an inlet pipe A and an outlet pipe B.

P10. • Inlet pipe A adds water at a rate of 15 liters per minute when open.

- Outlet pipe B drains water at a rate of 8 liters per minute when open.
- A computer-controlled system operates the reservoir in a repeating 10-minute cycle:
 - For the first 6 minutes, both Pipe A and Pipe B are kept open.
 - For the next 4 minutes, Pipe A is closed and Pipe B is kept open.

- (a) Express the net change in water volume during one complete 10-minute cycle as a signed integer.
- (b) If the reservoir initially contains 250 liters of water, write a mathematical expression using BODMAS to represent the volume of water in the reservoir after 35 minutes (which is exactly 3.5 complete cycles). Simplify the expression to find the final volume.
- (c) At what exact minute from the start of the first cycle does the reservoir reach its absolute maximum water volume, and what is that maximum volume?

SECTION C: COMPLETE SOLUTIONS KEY

Part 1: Multiple Choice Solutions

Q1. Answer: (a) 15**Step-by-step Evaluation:**

$$\begin{aligned}
 18 - [14 - \{15 - (8 - \overline{7 - 3})\}] &= 18 - [14 - \{15 - (8 - 4)\}] \quad (\text{Vinculum first}) \\
 &= 18 - [14 - \{15 - 4\}] \quad (\text{Parentheses}) \\
 &= 18 - [14 - 11] \quad (\text{Curly brackets}) \\
 &= 18 - 3 = \mathbf{15} \quad (\text{Final subtraction})
 \end{aligned}$$

Q2. Answer: (a) 42**Step-by-step Evaluation:** First, find x :

$$x = -12 \div \{(-3) + 1\} = -12 \div \{-2\} = \mathbf{6}$$

Next, find y :

$$y = (-15 \div -3) - (-2) = 5 + 2 = \mathbf{7}$$

Then, find $x \times y$:

$$x \times y = 6 \times 7 = \mathbf{42}$$

Q3. Answer: (a) 40**Step-by-step Evaluation:** Evaluate curly brackets first (multiplication inside curly brackets is done before subtraction):

$$\begin{aligned}
 (-50) \div \{15 - (-5) \times (-2)\} \times (-4) &= (-50) \div \{15 - 10\} \times (-4) \\
 &= (-50) \div \{5\} \times (-4)
 \end{aligned}$$

By BODMAS rules, perform division and multiplication from left to right:

$$\begin{aligned}
 (-50) \div 5 &= -10 \\
 -10 \times (-4) &= \mathbf{40}
 \end{aligned}$$

Q4. Answer: (a) -630 m**Step-by-step Evaluation:** Initial position of the submarine is -450 m. Diving at 15 m/min for 12 minutes represents a change of:

$$12 \times (-15 \text{ m}) = -180 \text{ m}$$

The new depth is:

$$-450 \text{ m} + (-180 \text{ m}) = \mathbf{-630 \text{ m}}$$

Q5. Answer: (a) 11**Step-by-step Evaluation:** Let w be the number of incorrect answers. Marks from 7 correct answers: $7 \times 4 = 28$ marks. Deduction from incorrect answers: $w \times (-3) = -3w$ marks. Set up the total score equation:

$$28 - 3w = -5 \implies -3w = -5 - 28 \implies -3w = -33 \implies w = \mathbf{11}$$

She made exactly 11 incorrect attempts.**Q6. Answer: (a) 45****Step-by-step Evaluation:** Profit from 16 notebooks: $16 \times \text{Rs. } 5 = \text{Rs. } 80$. Net balance = $-\text{Rs. } 10$ (overall loss). Let p be the number of pens sold. Each pen causes a loss of $\text{Rs. } 2$.

$$80 - 2p = -10 \implies -2p = -90 \implies p = \mathbf{45}$$

She sold exactly 45 pens.

Q7. Answer: (a) -33

Step-by-step Evaluation: Evaluate from the innermost bracket:

$$\begin{aligned}
 -24 + [-18 \div \{12 \div (4 - \overline{2 - 4})\}] &= -24 + [-18 \div \{12 \div (4 - (-2))\}] \\
 &= -24 + [-18 \div \{12 \div (4 + 2)\}] \\
 &= -24 + [-18 \div \{12 \div 6\}] \\
 &= -24 + [-18 \div 2] \\
 &= -24 + (-9) = \mathbf{-33}
 \end{aligned}$$

Q8. Answer: (a) -39°C

Step-by-step Evaluation: *Initial freezing point* : -18°C . The temperature drops by 3°C per minute for 7 minutes:

$$\Delta T = 7 \times (-3^\circ\text{C}) = -21^\circ\text{C}$$

The final temperature is:

$$-18^\circ\text{C} + (-21^\circ\text{C}) = \mathbf{-39^\circ\text{C}}$$

Q9. Answer: (a) -13

Step-by-step Evaluation (HOTS 9): Evaluate $x \otimes y$ where $x = -4$ and $y = 3$:

$$x \otimes y = (-4 \times 3) - (-4 + 3) \div (-1)$$

Following order of operations:

$$\text{Multiplication part: } a \times b = -4 \times 3 = -12$$

$$\text{Addition inside brackets: } a + b = -4 + 3 = -1$$

$$\text{Division part: } (a + b) \div (-1) = -1 \div (-1) = 1$$

Putting them together:

$$x \otimes y = -12 - (1) = \mathbf{-13}$$

Q10. Answer: (a) 1

Step-by-step Evaluation (HOTS 10): Substitute the given constants into the expression:

$$B - \{C \times (D - \overline{E - F})\} = -12$$

$$-6 - \{2 \times (-1 - (E - 5))\} = -12$$

Solve for E systematically:

$$-6 - 2(-1 - E + 5) = -12$$

$$-6 - 2(4 - E) = -12$$

$$-6 - 8 + 2E = -12$$

$$-14 + 2E = -12$$

$$2E = -12 + 14$$

$$2E = 2 \implies E = \mathbf{1}$$

Part 2: Subjective Solutions**P1. Solution: 65**

Evaluate the innermost brackets and vinculum first:

$$\overline{4 - 6} = -2$$

The expression becomes:

$$(-15) \times (-4) - [-25 \div \{10 - (3 - (-2))\}]$$

Simplify round brackets:

$$3 - (-2) = 3 + 2 = 5$$

The expression becomes:

$$(-15) \times (-4) - [-25 \div \{10 - 5\}]$$

Simplify curly brackets:

$$10 - 5 = 5$$

The expression becomes:

$$(-15) \times (-4) - [-25 \div 5]$$

Simplify division inside square brackets:

$$-25 \div 5 = -5$$

Perform the outer multiplication and the final subtraction:

$$(-15) \times (-4) = 60$$

$$60 - (-5) = 60 + 5 = \mathbf{65}$$

P2. Solution: -48

Evaluate the vinculum first:

$$\overline{12 - 5} = 7$$

The expression becomes:

$$-36 + [-48 \div \{-12 + 2 \times (15 - 7)\}]$$

Simplify round brackets:

$$15 - 7 = 8$$

The expression becomes:

$$-36 + [-48 \div \{-12 + 2 \times 8\}]$$

Within the curly brackets, multiply first before adding:

$$2 \times 8 = 16 \implies -12 + 16 = 4$$

The expression becomes:

$$-36 + [-48 \div 4]$$

Evaluate the square brackets:

$$-48 \div 4 = -12$$

Perform final addition:

$$-36 + (-12) = \mathbf{-48}$$

P3. Solution: 58

Evaluate the components individually:

- Modulus part: $|-15 - (-3)| = |-15 + 3| = |-12| = 12$

- Division part inside round brackets: $(-24) \div (-6) = 4$
- Inner part of square brackets: $13 + (-3) \times 3 = 13 - 9 = 4$
- Division inside square brackets: $-40 \div 4 = -10$

Substitute these values back into the expression:

$$12 \times 4 - [-10]$$

Multiply first, then subtract:

$$48 - (-10) = 48 + 10 = \mathbf{58}$$

P4. Solution: 3°C

Analysis: *The explorer starts at an altitude of 0 m with a temperature of 15°C .* Find the final campsite altitude coordinate:

$$h_{\text{final}} = +2400 \text{ m} - 600 \text{ m} = +1800 \text{ m}$$

Calculate the total temperature drop. For every 300 m, the temperature decreases by 2°C :

$$\text{Intervals} = \frac{1800 \text{ m}}{300 \text{ m}} = 6$$

$$\text{Total Decrease} = 6 \times (-2^{\circ}\text{C}) = -12^{\circ}\text{C}$$

Calculate final temperature:

$$T_{\text{campsite}} = 15^{\circ}\text{C} + (-12^{\circ}\text{C}) = \mathbf{3^{\circ}\text{C}}$$

P5. Solution: Rs. 11,200

Write the financial transaction values as signed integers:

$$\text{Initial Balance} = +12500$$

$$\text{Checks Deposited} = +3 \times 4200 = +12600$$

$$\text{Salary Withdrawal} = -15600$$

$$\text{Balance Penalty} = -150$$

$$\text{Returned Purchase Credit} = +1850$$

Now, sum these coordinates chronologically:

$$\begin{aligned} \text{Balance} &= 12500 + 12600 - 15600 - 150 + 1850 \\ &= 25100 - 15600 - 150 + 1850 \\ &= 9500 - 150 + 1850 \\ &= 9350 + 1850 = \mathbf{11200} \end{aligned}$$

The firm's final bank balance is **Rs. 11,200**.

P6. Solution: (a) 1700 m, (b) 1970 m

Let sea level be 0.

(a) **Initial altitude of helicopter:** $h_1 = +1250 \text{ m}$.

Initial depth of submarine: $s_1 = -450 \text{ m}$.

Vertical distance:

$$D_1 = h_1 - s_1 = 1250 - (-450) = 1250 + 450 = \mathbf{1700 \text{ m}}$$

(b) **New helicopter altitude:** $h_2 = 1250 + 150 = +1400 \text{ m}$.

New submarine depth: $s_2 = -450 - 120 = -570 \text{ m}$.

New vertical distance:

$$D_2 = h_2 - s_2 = 1400 - (-570) = 1400 + 570 = \mathbf{1970 \text{ m}}$$

P7. Solution: Correct = 26, Incorrect = 6

Analysis: Total questions = 40. Unattempted questions = $40 - 32 = 8$. Net deduction for unattempted questions: $8 \times (-1) = -8$ marks. Let c be the number of correct answers. Then incorrect answers $w = 32 - c$. Set up the net score equation:

$$5c - 2w - 8 = 110$$

Substitute $w = 32 - c$:

$$5c - 2(32 - c) - 8 = 110$$

$$5c - 64 + 2c - 8 = 110$$

$$7c - 72 = 110$$

$$7c = 110 + 72$$

$$7c = 182 \implies c = \mathbf{26}$$

Find the number of incorrect answers:

$$w = 32 - 26 = \mathbf{6}$$

Ananya had 26 correct and 6 incorrect answers.

P8. Solution: (a) 400 chips, (b) 600 chips

(a) Let the number of Grade-B chips sold be x . Profit from 450 Grade-A chips: $450 \times \text{Rs. } 120 = \text{Rs. } 54,000$. Loss from x Grade-B chips: $x \times \text{Rs. } (-45) = -45x$. Set up the net profit equation:

$$54000 - 45x = 36000 \implies -45x = 36000 - 54000$$

$$-45x = -18000 \implies x = \frac{-18000}{-45} = \mathbf{400}$$

The dealer sold exactly 400 Grade-B chips.

(b) Let the minimum number of Grade-A chips needed be y . Loss from 600 Grade-B chips: $600 \times \text{Rs. } (-45) = -\text{Rs. } 27,000$. We want the net profit to be at least Rs. 45,000:

$$120y - 27000 \geq 45000$$

$$120y \geq 45000 + 27000 \implies 120y \geq 72000$$

$$y \geq \frac{72000}{120} \implies y \geq \mathbf{600}$$

The dealer must sell at least 600 Grade-A chips.

P9. Solution: 12 distinct coordinate pairs

Step 1: Simplify the algebraic denominator first.

In the denominator:

$$D(x, y) = y - (x + \overline{y - 2})$$

Note that the bar is a grouping symbol equivalent to parentheses, so $\overline{y - 2} = y - 2$:

$$D(x, y) = y - (x + y - 2)$$

$$= y - x - y + 2$$

$$= \mathbf{2 - x}$$

Notice that the variable y is entirely eliminated from the denominator! Therefore, the function simplifies generally to:

$$f(x, y) = [x \div (2 - x)]$$

Step 2: Determine when the division is defined and evaluates to an integer.

For $f(x, y)$ to be defined, the denominator $2 - x \neq 0 \implies x \neq 2$. For $f(x, y)$ to be an integer, the term $(2 - x)$ must be an integer divisor of x . Let's test all possible integers in our domain $-3 \leq x \leq 3$:

- $x = -3$: $2 - x = 5 \implies -3/5$ (not an integer).
- $x = -2$: $2 - x = 4 \implies -2/4 = -1/2$ (not an integer).
- $x = -1$: $2 - x = 3 \implies -1/3$ (not an integer).
- $x = 0$: $2 - x = 2 \implies 0/2 = 0$ (Integer! Valid for all y).
- $x = 1$: $2 - x = 1 \implies 1/1 = 1$ (Integer! Valid for all y).
- $x = 2$: $2 - x = 0$ (Undefined).
- $x = 3$: $2 - x = -1 \implies 3/(-1) = -3$ (Integer! Valid for all y).

Thus, x can take exactly three values: $x \in \{0, 1, 3\}$. Since y does not affect the denominator or numerator, for each valid x , we can pair it with any of our allowed y -values: $y \in \{1, 2, 3, 4\}$. This yields exactly $3 \times 4 = 12$ distinct coordinate pairs:

$$(x, y) \in \{(0, 1), (0, 2), (0, 3), (0, 4), (1, 1), (1, 2), (1, 3), (1, 4), (3, 1), (3, 2), (3, 3), (3, 4)\}$$

P10. Solution: (a) +10 liters, (b) 315 liters, (c) 6 minutes (292 liters)

(a) **Cycle Change:** During the first 6 minutes, both A and B are open:

$$\text{Rate} = +15 - 8 = +7 \text{ l/min} \implies \text{Volume Change} = 6 \times (+7) = +42 \text{ liters}$$

During the next 4 minutes, only B is open:

$$\text{Rate} = -8 \text{ l/min} \implies \text{Volume Change} = 4 \times (-8) = -32 \text{ liters}$$

Net cycle change:

$$\Delta V = +42 + (-32) = +10 \text{ liters}$$

(b) **BODMAS Expression for 35 minutes:** Since 35 minutes corresponds to 3 full cycles of 10 minutes, plus an additional 5 minutes of a fourth cycle: During the final 5 minutes, since $5 \leq 6$, both Pipe A and B remain open (rate = +7 l/min). Let's write this complete evaluation as a BODMAS expression:

$$V_{35} = 250 + 3 \times \{(15 - 8) \times 6 + (-8) \times 4\} + 5 \times (15 - 8)$$

Simplify inside curly brackets first:

$$\begin{aligned} V_{35} &= 250 + 3 \times \{7 \times 6 + (-32)\} + 5 \times (7) \\ &= 250 + 3 \times \{42 - 32\} + 35 \\ &= 250 + 3 \times \{10\} + 35 \\ &= 250 + 30 + 35 = \mathbf{315 \text{ liters}} \end{aligned}$$

The volume is exactly 315 liters.

(c) **Absolute Maximum Volume Point:** During any given 10-minute cycle, the volume rises at +7 l/min for the first 6 minutes, and then falls at -8 l/min for the remaining 4 minutes. Thus, the peak volume of any cycle occurs at exactly the 6-minute mark. For the first cycle, the maximum volume achieved is:

$$V_{\max} = 250 + 6 \times (15 - 8) = 250 + 6 \times 7 = 250 + 42 = \mathbf{292 \text{ liters}}$$